

Take Home Project #2 (25 Mark)

IS6751 Text & Web Mining

Due on 30 April 2025

1. Build Sentiment Classifiers with the following requirements.

1.1. Dataset

- Use **reviews_with_splits_lite.csv** in rnn-transformer-sentiment\data\rt-polarity: Sentence-Level dataset from Rotten Tomatoes (<http://www.rottentomatoes.com/>): 5331 positive sentences and 5331 negative sentences
- **Do not change split values in the file** since everyone should use the same test data.

1.2. Algorithm

- Use following code as baseline models.
 - Sentiment-Classification-with-RNNs-v3-revised.ipynb
 - Sentiment-Classification-with-transformer-models-v1.ipynb (use a gpu machine, such as google colab)

1.3. Tasks

- **Task 1:** Optimize various RNN models (using Sentiment-Classification-with-RNNs-v3-revised.ipynb) to develop the best performing model.
 - For Task 1, you are welcome to use your own RNN model code instead of the provided code (i.e., *Sentiment-Classification-with-RNNs-v3-revised.ipynb*), if you prefer.
- **Task 2:** Optimize various transformer-based models, such as BERT, DistilRoBERTa, and RoBERTa (using Sentiment-Classification-with-transformer-models-v1.ipynb), to develop the best performing model.
 - For Task 2, you are welcome to use your own Transformer-based model code instead of the provided code (i.e., *Sentiment-Classification-with-transformer-models-v1.ipynb*), if you prefer. Large Language Models can be used also.
 - For Task 2, even though I recommend you use a gpu machine, if you cannot use it with some reasons, such as usage limits in google colab, you may use the CNN model instead.

1.4. Improve the model using following approaches.

- Text pre-processing, such as removing stop words, and using lemmatization, case folding, etc.
- Change hyperparameters, such as learning rate, # of hidden units, mini-batch size, # of layers, dropout, batch norm, regularization, etc.
- Any other techniques if you like to use.
- Note that you cannot use existing models without training with the provided dataset. In other words, you need to train models with the provided dataset.

Submission:

Submit **one zip file** (use only zip compression file), named **home-project-no2-yourname.zip**, that contains **your report file, Jupyter Notebook file for each task, data files (i.e., input data) and the best model files (e.g., model.pth files)** through Turnitin on the class website.

- Write a report in Word or PDF that discusses your observations, such as test results with various approaches. The report should contain **up to 2,000 words** and **write down the total number of words on your cover page**.
- The report file should have a cover page.
- The Jupyter notebook files must show all output results of your Python code. So please make sure that you run all the cells in the notebook files before your submission.
- If model files are too big for Turnitin submission, you do not need to include them in your zip file.

- Note that Turnitin does not allow you to resubmit your assignment file.
- **Reports and required files submitted in after the due date will not be marked because of the strict university deadline.**